1

$$\cos x = (2 - \operatorname{tg} x)(1 + \operatorname{sen} x)$$

$$\cos x = \left(2 - \frac{\operatorname{sen} x}{\cos x}\right)(1 + \operatorname{sen} x)$$

$$\cos x = \frac{(2\cos x - \operatorname{sen} x)}{\cos x}(1 + \operatorname{sen} x)$$

$$\cos^2 x = 2\cos x + 2\cos x \operatorname{sen} x - \operatorname{sen} x - \operatorname{sen}^2 x$$

$$1 - \operatorname{sen}^2 x = 2\cos x + 2\cos x \operatorname{sen} x - \operatorname{sen} x - \operatorname{sen}^2 x$$

$$2\cos x + 2\cos x \operatorname{sen} x - \operatorname{sen} x - 1 = 0$$

$$2\cos x(1 + \operatorname{sen} x) - (\operatorname{sen} x + 1) = 0$$

$$(1 + \operatorname{sen} x)(2\cos x - 1) = 0$$

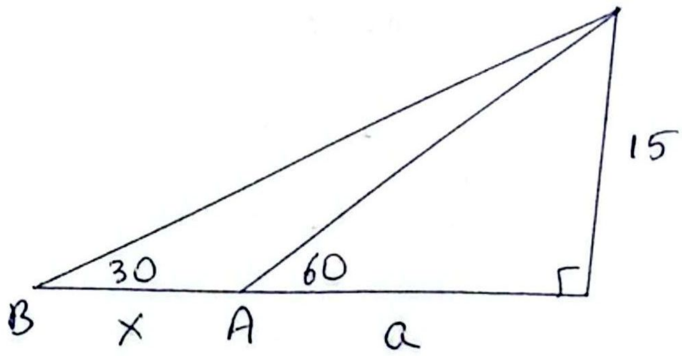
$$\operatorname{sen} x = -1 \longrightarrow x_0 = 270^\circ$$

$$\cos x = \frac{1}{2} \Longrightarrow x_1 = 60, \quad x_2 = 300$$

Luego

$$E = x_0 + x_1 + x_2 = 270 + 60 + 300 = 630^\circ$$

(2)



$$tg\,60 = \frac{15}{a} \Rightarrow a = \frac{15}{tg\,60} = \frac{15}{\sqrt{3}} = \frac{15\sqrt{3}}{\sqrt{3}\sqrt{3}} = \frac{15\sqrt{3}}{3}$$

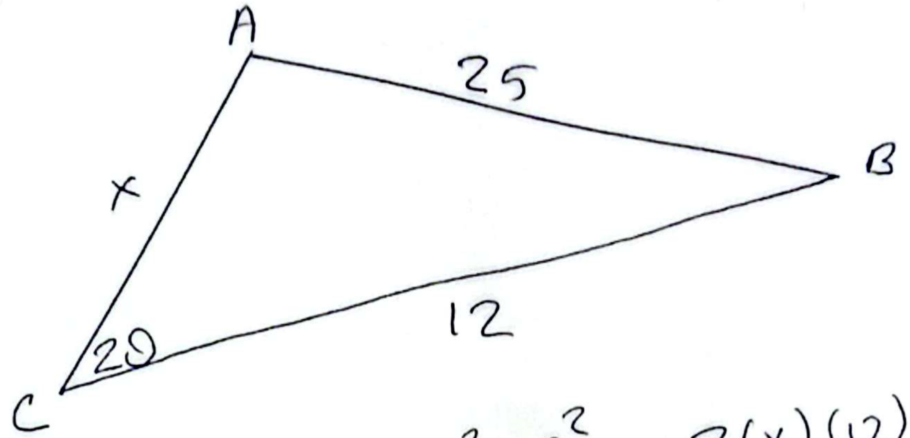
$$a = 5\sqrt{3}.$$

$$tg\,30 = \frac{15}{x+a} \Rightarrow x+a = \frac{15}{tg\,30}$$

$$x + 5\sqrt{3} = \frac{15}{\frac{1}{\sqrt{3}}} = 15\sqrt{3}$$

$$x = 10\sqrt{3}$$

(3)



$$25^2 = x^2 + 12^2 - 2(x)(12)\cos 20$$

$$625 = x^2 + 144 - 22{,}56x$$

$$x^2 - 22{,}56x - 481 = 0$$

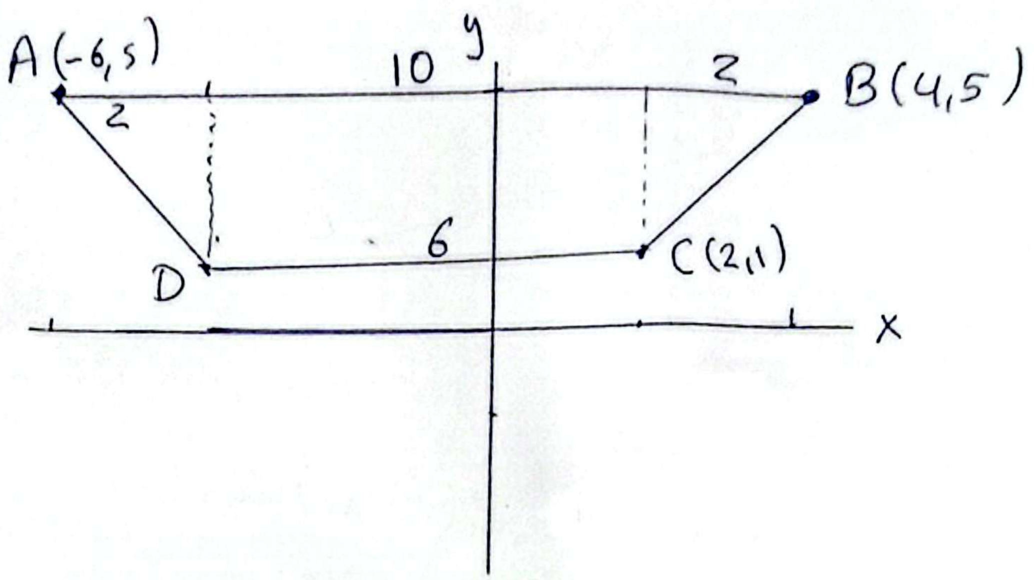
$$x = \frac{22{,}56 \pm \sqrt{(22{,}56)^2 - 4(1)(-481)}}{2(1)}$$

$$x = \frac{22{,}56 \pm 49{,}32}{2}$$

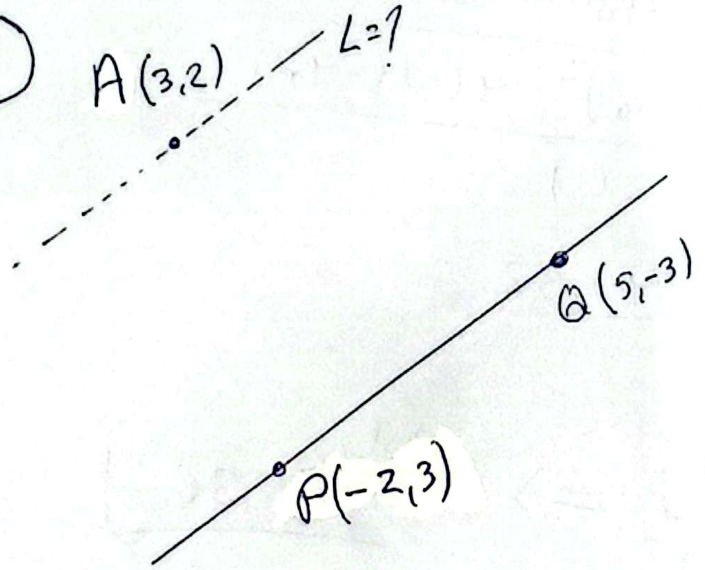
$$x_1 = 35{,}94 \Rightarrow$$

$$x_2 = -13{,}38$$

Sol. $x = 36$

(4) 

Base menor: $\overline{DC} = 6$

(5) 

$$m_{PQ} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-3-3}{5-(-2)} = -\frac{6}{7}$$

Como $L \parallel \overline{PQ} \Rightarrow L: \; y - y_0 = m(x - x_0)$

$$y - 2 = -\frac{6}{7}(x-3)$$

$$7y - 14 = -6x + 18$$

$$y = -\frac{6}{7}x + \frac{32}{7}$$